Explainable Machine Learning for Sports Injury Risk Analysis

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Research Objective

Rising Number of Injuries

- Intensification and professionalization
- Resulting in physical, psychological and economic damages

The "Black-Box" Barrier

- High ML Performance but lack transparency [1]
- Clinical problem

Objective

Develop an interactive framework for Bridging the Gap between Machine Learning and Sport Injury Risk Analysis by using new Explainable Artificial Intelligence (XAI) techniques.

Outline

Dataset used and Notations

University Football Injury Prediction Dataset [2]

- **800 Players** ($n = 800$ observations).
- **18 Features** ($p = 18$): Players Info (Age, Weight,...), Physical Abilities (Knee Strength,...), Life Habits (Stress Score, Nutrition Score...) at the beginning of the season.
- **Target Variable ($Y \in \{0, 1\}$):** Injury (1) vs. No Injury (0).
  Definition: Player missed ≥ 1 game next season due to a physical problem.

Notations

$$\mathbf{X} = \left[\begin{array}{cccc|c} x_1^{(1)} & x_1^{(2)} & \cdots & x_1^{(p)} & Y_1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_i^{(1)} & x_i^{(2)} & \cdots & x_i^{(p)} & Y_i \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_n^{(1)} & x_n^{(2)} & \cdots & x_n^{(p)} & Y_n \end{array}\right]$$

- $j \in [\![1, p]\!], x^{(j)}$ is the j-th feature.
- $i \in [\![1, n]\!], x_i$ is the i-th observation.

Logistic Regression Model

Logistic Regression known as a **Generalized Linear Model**

Model Definition

$$\begin{cases} Y_i \sim \mathcal{B}(\pi(x_i)) \\ logit(\pi(x_i)) = \ln\left(\frac{\pi(x_i)}{1-\pi(x_i)}\right) = \beta_0 + \beta_1 x_i^{(1)} + ... + \beta_p x_i^{(p)} \end{cases}$$

Where $\forall j \in [\![1, p]\!], \beta_j \in \mathbb{R}$

Estimation of the parameters using the Fisher-Scoring Algorithm to minimize the Log-Loss function:

$$J(\underline{y}, \beta) = -\frac{1}{n} \sum_{i=1}^{n} [y_i \log(\pi_i) + (1 - y_i)\log(1 - \pi_i)]$$

Where $\underline{y} = (Y_1, ..., Y_m)$ *and m is the number of observation in the testing subset*

Logistic Regression Model Interpretation

Intrinsic Interpretability

The linear equation ensures a direct and transparent interpretation.

After the parameter estimation:

$$\text{logit}(\hat{\pi}(x_0)) = \ln\left(\frac{\hat{\pi}(x_0)}{1-\hat{\pi}(x_0)}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_0^{(1)} + \cdots + \hat{\beta}_p x_0^{(p)}$$

Global Interpretation

Magnitude: $|\hat{\beta}_j|$ indicates feature importance.

Odds Ratios: quantify the effect.

Local Interpretation

Contribution: $\hat{\beta}_j \times x_i^{(j)}$ for a specific player i.

Random Forest Architecture & Performance

Model Architecture: Ensemble Learning [3]

- Aggregates multiple **Decision Trees** (binary flowcharts).
- Captures complex, non-linear biomechanical relationships.
- *Prediction:* Final outcome determined by majority vote.

Model Characteristics (*via k-fold cross-validation*)

- Number of trees: 200
- Maximum depth: 5
- Features per split: $\sqrt{p}$

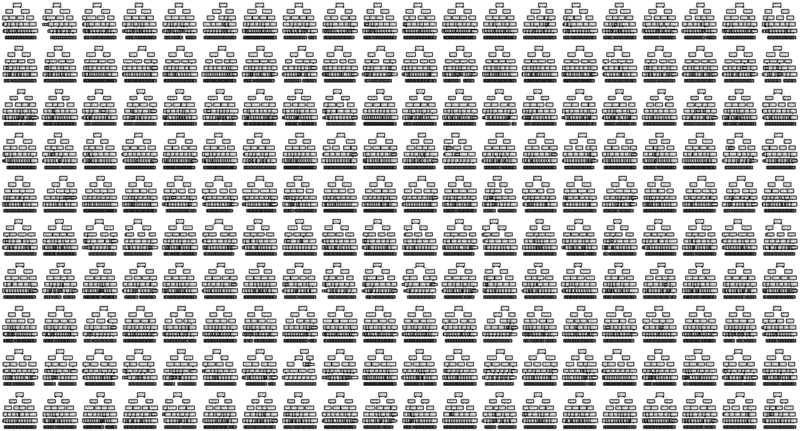
Performance on Testing Set - Accuracy: **94.3%**

	Pred: Safe	**Pred: At Risk**
True: Safe	[77]	[3]
True: At Risk	[6]	[74]

Black-Box Problem

Black Box Problem

It is practically impossible to interpret 200 combined trees and identify impactful features.



Clinical Application: The Risk Grid

The Limitation of Binary Classification

- Categorizing players strictly as "At Risk" or "Safe" lacks clinical nuance.
- *Example:* A player with a 90% injury probability requires a different intervention than a player with 55%.

Proposed Solution: Nuanced Risk Grid

- Extract the **output probability** rather than the final binary class.

$$\hat{\pi}(x_0) = \frac{1}{B}\sum_{b=1}^{B} h_b(x_0) \text{ where, } B = 200 \text{ and } h_b \text{ is the prediction of the b-th tree.}$$

- Group players into specific risk tiers to help medical staff make tailored, data-driven decisions.

Cross-Tabulation Matrix: Injury vs. Risk Grid

	Risk Grid				
True Injury	**1 (0-10%)**	**2 (10-40%)**	**3 (40-60%)**	**4 (60-90%)**	**5 (90-100%)**
0	15	50	15	0	0
1	0	1	15	47	17

Global Interpretability: Partial Dependence & Individual Conditional Expectations Curves

Methodology

ICE [7]: Evolution of injury probability for an *individual* player when varying one feature (others frozen).

$$f_j(x_i^{(j)}) = \int \hat{f}(x_i^{(j)}, x_C)\, d\mathbb{P}(x_C)$$

PD [6]: The *average* trend of all ICE curves.

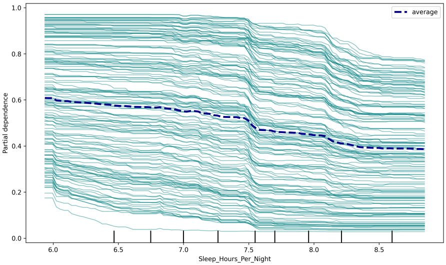
$$f_j(x^{(j)}) = \int \hat{f}(x^{(j)}, x_C)\, d\mathbb{P}(x_C)$$

Where $X_C = X \setminus \{x^{(j)}\}$.

Risk-Increasing: Stress Level, Reaction Time.

Protective: Sleep Hours, Balance Test.

Focus: Sleep Hours per Night parameter



Clinical Insight

- **-20% injury risk** (6.5h $\rightarrow$ 8.5h).
- Key thresholds at **7.5h** and **8.3h**.

Local Interpretability: SHAP Values

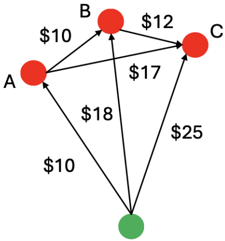
Cooperative Game Theory: **Shapley Values** is a method to fairly distribute a total payout among n contributors based on their marginal contributions **[4]**.

The Shapley Equation:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n-|S|-1)!}{n!}(v(S \cup \{i\}) - v(S))$$

Where:

S is a subset of contributors (features)
$v(S)$ is the final payout (prediction) of subset S.



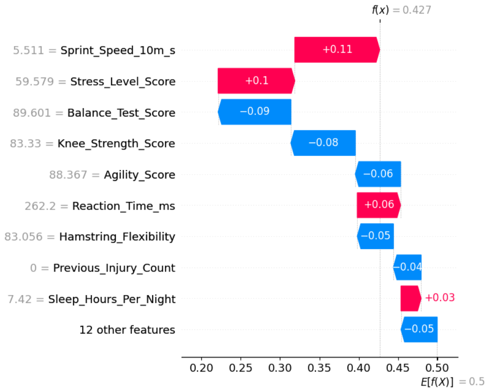
Application to Machine Learning (SHAP, 2017) [5]

Our Clinical Context

- **The "Taxi Fare"** = The final injury risk prediction.
- **The "Passengers"** = The player's specific biomechanical and psychological metrics.

Local Interpretability: Individual Player Profile

Individual SHAP Explanation



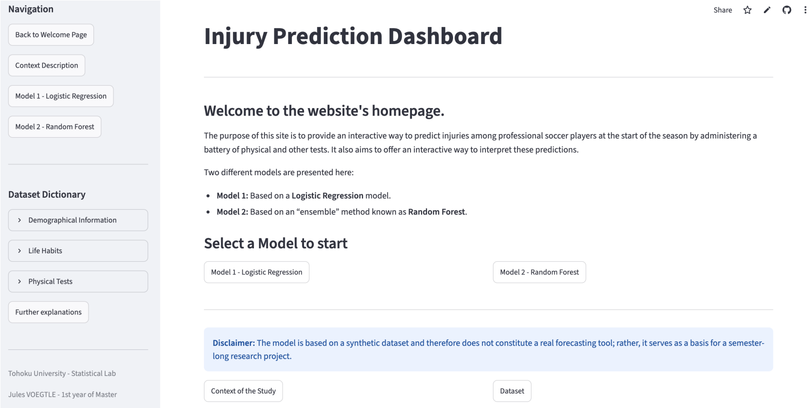
Risk Grid Placement

- **Base Value (Average):** 50.0%
- **Individual Prediction:** 42.7%
- **Assigned Tier:** Class 3 (40-60%)

Key Drivers (Feature Impact)

- **Protective Factors (↓):** Balance Test Score, Knee Strenght Score.
- **Features to improve:** Sprint Speed, Stress Level

Web Interface



Interactive Dashboard: https://sport-injury-risk-analysis.streamlit.app/

Conclusion & Acknowledgments

Project Achievements

- **Performance vs. Transparency:** XAI bridges the gap, allowing the Random Forest to match the interpretability of the Baseline Model.
- **Clinical Superiority:** Outperforms traditional methods by identifying non-linear thresholds (e.g., Sleep Hours).
- **Actionable Insights:** Provides both global trends and highly localized metrics for individualized player care.

Thank you for your attention.

References

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